https://labex.io/labs

Basics of the Shell

pwd – Print Working Directory

What it does: Shows the current directory path.

Example:

pwd

Output: /home/user

ls – List Files & Directories

What it does: Displays contents of a directory.

Example:

ls -l

Output: Shows files with permissions, size, and dates.

cd – Change Directory

What it does: Move between directories.

Example:

cd /etc

clear – Clear Terminal Screen
Example:

clear

man – Manual Pages

What it does: Shows help documentation for a command

Example:

man ls

history – Command History

Example:

history | grep ls

alias – Create Shortcuts

Example:

touch

File and Directory Management

touch – Create Empty File

Example:

touch notes.txt**cat – View File Content**

Example:

cat notes.txt

less – View Large Files

Example:

less /var/log/syslog

head / tail – View Start/End of File

Example:

head -n 5 notes.txt # First 5 lines
tail -n 10 notes.txt # Last 10 lines

mkdir – Create Directory

Example:

mkdir projects

cp – Copy Files
Example:

cp notes.txt backup/

mv – Move/Rename Files
Example:

mv notes.txt ideas.txt

rm – Remove Files

Example:

rm ideas.txt

find – Search for Files
Example:

find /home -name "*.txt"

File Permissions and Ownership

l – Change Permissions
Example:

Two ways to set permissions:

1. Symbolic mode (using letters)

- **u = user (owner)**

- **g = group**
- **o = others**
- **a = all (u+g+o)**
- **Operators:**
    - **+ → add permission**
    - **- → remove permission**
    - **= → set exact permission**

Example:

```
chmod u+x script.sh     # give owner execute permission
chmod g-w file.txt      # remove write permission from group
chmod a=r file.txt      # everyone gets read-only
```

2. Octal (numeric) mode

Each permission has a number:

- **r (read) = 4**
- **w (write) = 2**
- **x (execute) = 1**

Add them up for each group:

- **chmod 755 file.sh → rwxr-xr-x**
- **chmod 644 file.txt → rw-r--r--**
- **chmod 700 secret.txt → rwx------**

chmod 755 script.sh
Add execute permission for user
chmod u+x script.sh

Remove write permission from others
chmod o-w file.txt

Give read permission to everyone
chmod a+r notes.txt

chown – Change Owner

Example:

sudo chown user:group file.txt

Text Processing

echo – Display Text

Example:

echo "Hello World"

grep – Search Text

Example:

grep "error" logfile.log

awk – Extract Columns

Example:

awk '{print $1, $3}' data.txt

sed – Edit Text in Files

Example:

sed 's/old/new/g' file.txt

System Information

uname – OS Information

Example:

uname -a

df – Disk Usage

Example:

df -h

free – Memory Usage

Example:

free -h

uptime – System Uptime

Example:

uptime

Process Management

ps – Show Processes

Example:

ps aux | grep ssh

top – Real-Time Process Monitoring

Example:

top

kill – Terminate Process

Example:

kill 1234

Networking

ping – Test Network
Example:

ping -c 4 google.com

ssh – Remote Login
Example:

ssh user@server

scp – Secure Copy
Example:

scp file.txt user@remote:/path/

Archiving & Compression

tar – Archive Files
Example:

tar -czvf backup.tar.gz folder/

gzip / gunzip
Example:

gzip file.txt
gunzip file.txt.gz

User Management

passwd
Example:

passwd username

useradd

Example:

sudo useradd newuser

sudo

Example:

sudo apt update

3. Networking Commands

Command	Description
ping <host>	Check network connectivity
ifconfig / ip a	Show IP addresses
netstat -tuln	Show open ports
wget <url>	Download files from the web

Package Management (Ubuntu/Debian)

Command	Description
sudo apt update	Refresh package list

sudo apt upgrade	Upgrade packages
sudo apt install <pkg>	Install package
sudo apt remove <pkg>	Remove package